

CASE STUDY · AI QA AUTOMATION



QA Pilot

Give it a URL. It learns the app, writes the tests, runs them, and reports the bugs.

AI

QA Automation

Selenium

Python

Claude

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localhost:8000 · Run overview

THE CHALLENGE

Small teams ship web apps nobody really tested.

Manual QA is slow and expensive, test suites take weeks to write, and the bug reports that do arrive are vague.



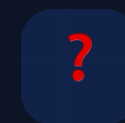
Slow and costly

Clicking through every role and every form before each release takes days — so it gets skipped.



Gaps nobody sees

Permission holes, wrong totals and broken state rules hide in screens that only some roles reach.



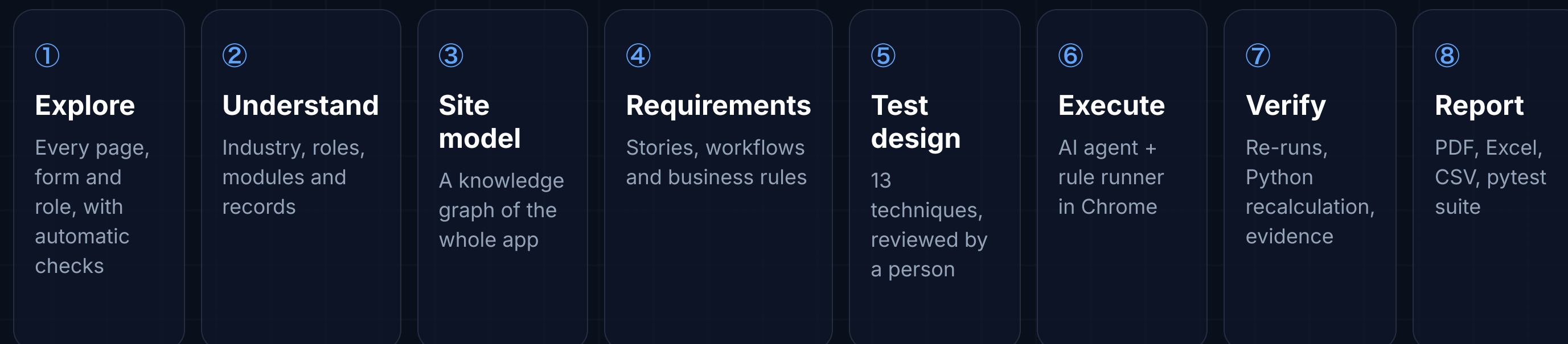
"It doesn't work"

Bug reports without steps, data or evidence cost developers hours to reproduce.

THE SOLUTION

An AI QA engineer that runs on your own computer.

QA Pilot explores the app like a new tester, writes down what it should do, tests it in a real browser and proves every bug it reports.



100% local · no database · your data stays in a folder · the only external call is the Claude API.

WORKS ON ANY INDUSTRY

Three very different apps. No hints given.

CarePoint Clinic

Healthcare · clinic
management

Insurance coverage on invoices, no double-booked doctors,
inactive doctors not bookable, cancelled visits can't be
completed

13/15

planted bugs found

DriveNow Rentals

Rental · car rental back
office

Minimum one-day rental, 10 % discount from 7 days, no
overlapping bookings, cars in maintenance not bookable

15/15

planted bugs found

StockRoom POS

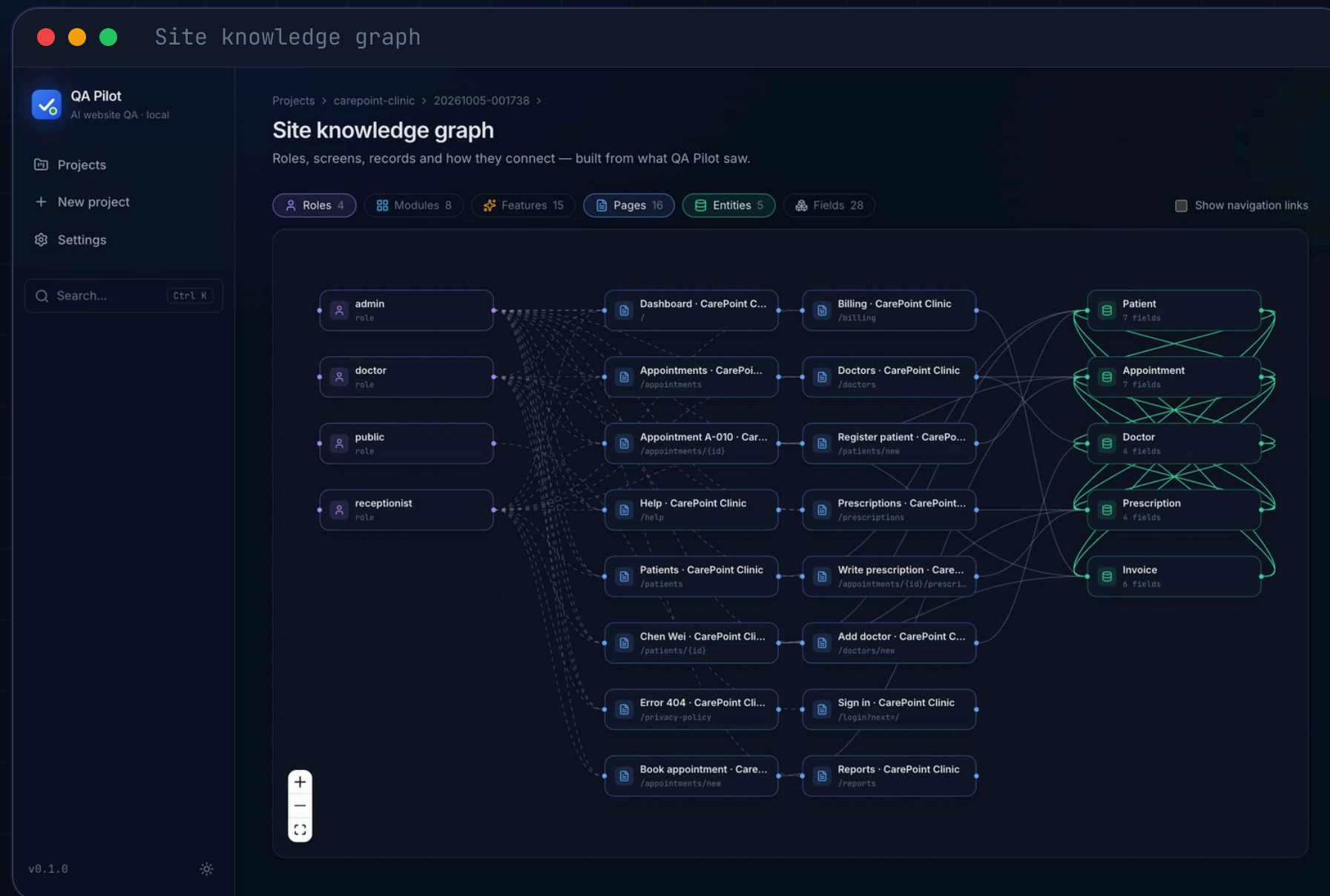
E-commerce · retail point
of sale

Total = subtotal + VAT, stock moves on sale and refund, one
refund per sale, profit report admin-only

13/15

planted bugs found

It builds a map of the app before it tests anything.



- ✓ Roles, screens, records and flows in one graph
- ✓ Which role reaches which screen — the base of the permission matrix
- ✓ Domain packs for 12 industries, plus Claude for the details
- ✓ Requirements and business rules a person confirms

Real test cases, with data and expected values.

- Smoke
- CRUD · E2E
- Validation
- Boundary values
- Equivalence classes
- State transitions
- Permissions
- Data integrity
- Business rules
- Responsive · Accessibility

Business rules come with computed results — *fee 120.00 at 50 % coverage → patient pays 60.00* — and are re-checked in Python after the run.

Test cases · review and approve

QA Pilot
AI website QA · local

Projects
+ New project
Settings

Search... Ctrl K

Projects > carepoint-clinic > 20261005-001738 >

Test cases

Review, edit and approve. Only **approved** cases run. 71 approved · 0 to review · 0 skipped.

Requirements Regenerate Add test

ESTIMATE FOR 71 APPROVED CASES 🕒 ≈ \$5.56 ⌚ ≈ 66.8 min Excel Gherkin 📄 📄 📄 Run approved tests

343 steps · claude-sonnet-5-5

Last Excel export: exports/test_cases.xlsx

Search... Technique: all Priority: all Status: all Select all shown Approve Skip

Access 1 select module

TC-ACC-001

Signed-out visitors cannot open protected screens
public

permission

P1

🛡️

approved

Accessibility 3 select module

TC-ACC2-001

Accessibility scan of admin screens
admin

accessibility

P3

🛡️

approved

TC-ACC2-002

Accessibility scan of doctor screens
doctor

accessibility

P3

🛡️

approved

TC-ACC2-003

Accessibility scan of receptionist screens
receptionist

accessibility

P3

🛡️

approved

Appointment 15 select module

TC-APP-003

Doctor cannot open /appointments/new by direct URL
doctor

permission

P1

🛡️

approved

TC-APP-004

E2E appointment lifecycle: book, check in, complete, invoice, pay
receptionist · designed by Claude

e2e

P1

🕒

approved

TC-APP-005

State transition: scheduled to checked in is allowed
receptionist · designed by Claude

state transition

P1

🕒

approved

TC-APP-006

State transition: scheduled to cancelled is allowed

state transition

P1

🕒

approved

Watch it test, step by step.

QA Pilot

AI website QA · local

Projects

New project

Settings

Search...

Ctrl K

Projects > stockroom-pos >

Run 20261005-002356

running

full mode

39m 20s

\$5.361 · 681 AI calls

Site profile

Site model

Requirements

Reports

Review test cases

HTML report

Stop

96%

Explore

22 pages, 2 findings (public:0...)

38s

Understand

ecommerce · retail point of sale with...

15s

Site model

12 screens, 4 entities, 3 roles,...

0s

Requirements

14 stories, 3 workflows, 28...

28s

Test design

63 test cases ready for review

1m 2s

Execute

31 passed · 28 failed · 4 blocked...

17m 41s

Verify

Re-run 2/3 of TC-SAL-004

Report

Exploration report written

0s

Live browser

TC-UIX-001

StockRoom POS

Shop back office

Dashboard

New sale

Sales

Products

Purchases

Profit report

Sale 5-00001

Nadia Hussain Admin Log out

2026-10-02 · cashier Ben Turner · paid by card · completed

ITEM	QTY	UNIT PRICE	LINE TOTAL
Sparkling Water 1L	6	\$1.20	\$7.20
Dark Chocolate Bar	2	\$2.20	\$4.40
Subtotal		\$11.60	
VAT (15%)		\$1.74	
Total		\$13.34	

Refund sale

What QA Pilot is doing

4 x 7 0

17:30:35

Re-run 2/3 of TC-SAL-004

17:30:30

Re-run 3/3 of TC-REP-001

17:30:25

Re-run 2/3 of TC-REP-001

17:29:31

Re-run 3/3 of TC-PUR-009

17:28:06

Re-run 2/3 of TC-PUR-009

17:26:47

Re-run 3/3 of TC-PUR-008

17:25:17

Re-run 2/3 of TC-PUR-008

17:24:48

Re-run 3/3 of TC-PRO-021

17:24:19

Re-run 2/3 of TC-PRO-021

17:24:08

Re-run 3/3 of TC-PRO-008

17:23:57

Re-run 2/3 of TC-PRO-008

17:23:46

Re-run 3/3 of TC-PRO-005

17:23:34

Re-run 2/3 of TC-PRO-005

17:23:26

Re-run 2/3 of TC-PRO-005

Proven, reproducible, with the evidence attached.

QA Pilot
AI website QA - local

Projects

+ New project

Settings

Search...

Ctrl K

Projects > carepoint-clinic > 20261005-001738 > Bugs >

BUG-001

Doctor can open restricted page /doctors/new by direct URL

critical

P1

New

reproduced 3/3

confidence 92%

Replay in browser

PDF

Confirm

Mark fixed

Not a bug

CarePoint Clinic

Clinic management

Dashboard

Patients

Appointments

Prescriptions

Add doctor

Dr. Daniel Lee

Doctor

Logout

Name

Specialty

General practice

Consultation fee (\$)

Save doctor

BUG-001: expected Access is refused (403, redirect or error) for this role. — actual The page opened for doctor: HTTP 200, title "Add doctor - CarePoint Clinic", headings: Add doctor Also...

BUG CLIP

0:00

FULL TEST VIDEO

0:00

/doctors/new is not in the doctor menu, but typing its URL opens it with full content.

EXPECTED

Access is refused (403, redirect or error) for this role.

ACTUAL

The page opened for doctor: HTTP 200, title "Add doctor - CarePoint Clinic", headings: Add doctor Also reproduced as receptionist.

PRECONDITIONS

Logged in as doctor

STEPS TO REPRODUCE

1. Log in as doctor

2. Open /doctors/new by typing its URL in the address bar

Severity reasoning: Broken access control: a role reaches a screen meant for other roles that exposes privileged data or actions.

Environment

Module

Doctor

Category

permission

Role

doctor

URL

http://localhost:8101/doctors/new

Browser

Chrome 154.0.8037.93

OS

Darwin 25.5.0

Viewport

1440x900 (desktop)

v0.1.0

- Reproduced 3/3 · confidence
- Annotated screenshot
- Expected vs actual
- Steps with real data
- 10–25 s clip + full video
- One-click replay

QA Pilot

08 / 12

Built to be trusted.

Agent loop

One action per Claude turn, an indexed element list plus screenshot, working memory and loop detection.

Self-healing replay

Every agent run becomes a replay script with multi-strategy locators; broken locators hand back to the agent.

Re-run verification

Failures run again: 3/3 confirmed, 1/3 flaky → needs review. Rule numbers re-computed in Python.

Permission matrix

Expected access from each role's menus vs what direct URLs really allow — holes in red.

Privacy & safety

Safe Mode guard on every click, encrypted credentials, PII blur, authorization record.

Local, file-based

No database: atomic JSON writes, file locks, schema versions — zip a project and move it.

41 of 45 planted bugs found.

91%

detection rate across three demo apps (41/45)

9

false positives in 67 bug reports

\$16.84

total Claude cost for all three apps, end to end

CarePoint Clinic

13/15 planted · 7 other genuine issues · 2 false positives

71 tests · \$5.88

DriveNow Rentals

15/15 planted · 2 other genuine issues · 1 false positive

60 tests · \$4.87

StockRoom POS

13/15 planted · 4 other genuine issues · 6 false positives

63 tests · \$6.09

Planted bugs are listed in answer keys the engine never reads; each match was checked and is listed in docs/benchmarks.json.

Proven tools, put together carefully.

BROWSER

Selenium 4

PROTOCOL

Chrome DevTools

AI

Anthropic Claude

BACKEND

Python · FastAPI

LIVE UPDATES

Server-Sent Events

MODELS

Pydantic 2

GRAPH

NetworkX · React Flow

FRONTEND

React 19 · TypeScript

STYLING

Tailwind · shadcn/ui

ACCESSIBILITY

axe-core

REPORTS

Chrome PDF · openpyxl

VIDEO

ffmpeg

TESTS

pytest · Vitest

QUALITY

Ruff · mypy · CI

STORAGE

JSON files, no DB

PLATFORMS

Windows · macOS · Linux



Want automated QA for your web app?

I build QA automation and custom AI testing tools — from Selenium suites to engines like this one.

Muzahidul Rahman

QA Automation Engineer & AI Automation Developer

[Hire me on Upwork](#)

github.com/YOUR_USERNAME/qa-pilot